

Access Redis and PostgreSQL Databases with Haskell



Named after logician Haskell Curry, Haskell is a standardised, general-purpose, purely functional programming language, with non-strict semantics and strong static typing. This tenth article on Haskell explores access to Redis and PostgreSQL databases using Haskell modules.

The Hackage website at <https://hackage.haskell.org/packages/#cat:Database> provides a vast number of database packages that you can use, a couple of which will be covered here.

You will need to install the *cabal-install* tool on Fedora, for example, using the following command:

```
$ sudo yum install cabal-install
```

Connecting to the Redis database

Let's use the *hedis* package to connect to the Redis server. Install the Fedora dependency package *alex*, and the Redis server as shown below:

```
$ sudo yum install alex redis
```

You can then install the *hedis* package using the following commands:

```
$ cabal update
```

```
$ cabal install hedis
```

This installs the latest *hedis* version 0.6.5. You can now start the Redis server on Fedora using the *service* command:

```
$ sudo service redis start
```

You can then test connectivity to the Redis server using the *redis-cli* command by issuing the *PING* command as follows:

```
$ redis-cli
```

```
127.0.0.1:6379> PING
PONG
```

You can also test the same using the *hedis* package inside the GHCi interpreter as illustrated below:

```
$ ghci
```